

Jinwon Kim

Robotics Engineer

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SUMMARY

- Part-time PhD researcher (company-sponsored, Korea University) on VLM-based robot autonomy and RL for legged locomotion — building sim-to-real pipelines on FuRO with MuJoCo, mjlab, and CMA-ES-based parameter identification (PACE).
- Shipped end-to-end autonomy stack on the Vision 60 quadruped robot (SLAM/GLIM, DLIO localization, Nav2+MPPI planning, Behavior Tree missions) as Robot-Intelligence Team Lead at KRM.
- 4 government-funded R&D programs (IITP, NIPA, KOITA, defense cluster), 2 corresponding-author papers, 1 filed patent, and Ministry of National Defense Drone-Bot Challenge Grand Prize (2 consecutive years, 2023 & 2024).

EXPERIENCE

- Korea Robot Manufacturing** 🌐, Robotics Engineer (Robot-Intelligence Team Lead) May 2023 – present
3 years
- Led 5 engineers on autonomous navigation for quadruped robots (Vision 60)
 - Wrote and deployed the full autonomy stack on Jetson Orin with ROS 2, covering SLAM (GLIM), localization (DLIO), path planning (Nav2 + MPPI), and mission execution (Behavior Tree)
 - Designed ROUTE solution, a graph-based mission system with plugin architecture for patrol. Supports per-node tasks like PTZ surveillance, docking, and gait switching
 - Built PACE (Parameter Actuator Calibration Engine) for MuJoCo sim-to-real transfer. Joint dynamics error under 5% NRMSE via CMA-ES
 - 2 papers as corresponding author (ICCAS 2025, KRoC 2026); 1 patent filed on robot path planning
 - Contributed to 4 government-funded R&D projects (IITP, NIPA, KOITA, defense cluster)
- Korea Institute of Science and Technology** 🌐, Student Researcher May 2019 – Dec 2019
8 months
- Collected 1,000+ annotated images via web crawling for delivery robot object detection
 - Wrote real-time multi-object detection and tracking with YOLOv3 and Siamese network
 - Published at KRoC 2020; patent registered (KR-10-2020-0026298)

EDUCATION

- PhD** **Korea University**, Computer Science and Engineering Seoul, South Korea
Sept 2025 – present
- Research: VLM-based robot autonomy, reinforcement learning for quadruped locomotion
- MS** **Korea Advanced Institute of Science and Technology**, Robotics Daejeon, South Korea
Mar 2021 – Feb 2023
- Scalable Graphics, Vision, and Robotics Lab 🌐 (Advisor: Prof. Sung-eui Yoon)
- Focus: Reinforcement Learning, Deep Learning, Trajectory Optimization
- BS** **Kwangwoon University**, Division of Robotics Seoul, South Korea
Mar 2017 – Feb 2021
- Focus: Robot Control, Navigation, Computer Vision
 - GPA: 4.23 / 4.5

PUBLICATIONS

- Urban Autonomous Robot Navigation via Natural-Language Route Instructions and VLM-Based Situation Understanding** Jan 2026
- Minje Kim, Kyungtae Park, *Jinwon Kim*[†]
(Korea Robotics Society Annual Conference (KRoC 2026))

ATBT: Adaptive Topological Map-Based Behavior Tree for Quadruped Robots Woosung Yoon, Junhyeok Choi, Kyungtae Park, Joonghyun Shin, Seungtaek Sung, Jinwon Kim [†] (International Conference on Control, Automation and Systems (ICCAS 2025))	Jan 2025
Collision-Backpropagation based Obstacle Avoidance Method for a Legged Robot Expressed as a Simplified Dynamics Model Jinwon Kim , Heechan Shin, Sung-eui Yoon (International Conference on Control, Automation and Systems (ICCAS 2022), Busan, Korea)	Jan 2022
Collision Backpropagation-based Obstacle Avoidance Method for a Legged Robot with Simplified Dynamics Model Jinwon Kim , Heechan Shin, Sung-eui Yoon (Korea Robotics Society Annual Conference (KRoC 2022), Pyeongchang, Korea)	Jan 2022
Robust Multi Object Detection Using Siamese Network Jinwon Kim , KangGeon Kim (Korea Robotics Society Annual Conference (KRoC 2020), Pyeongchang, Korea)	Jan 2020

PATENTS

Robot Path Control Apparatus and Method for Determining Movement Path Considering Obstacles Jinwon Kim	2024
Robust Multi-object Detection Apparatus and Method Using Siamese Network KangGeon Kim, Jinwon Kim	2022

PROJECTS

Quadruped Robot Autonomous Navigation Full-Stack System Team lead — SLAM, localization, planning, and ROUTE mission system for Vision 60 - Ran GLIM SLAM, DLIO localization, Nav2+MPPI planning, and ROUTE missions on Jetson Orin with ROS 2 Jazzy - ROUTE plugin architecture supports per-edge cost functions and per-node mission actions (acoustic camera, detection, PTZ)	Korea Robot Manufacturing Dec 2024 – Jan 2026
Sim-to-Real Pipeline for Quadruped Robot (FuRo) MuJoCo simulation environment and PACE parameter identification for RL-based locomotion - PACE pipeline identifies 6 physics parameters (armature, damping, frictionloss, kp, kd, latency) from 1kHz real robot data using CMA-ES - Set up mlab-based RL training environment on top of PACE-calibrated dynamics for sim-to-real transfer	Korea Robot Manufacturing Dec 2025 – present
Quadruped Robot System for Defense Monitoring and Reconnaissance National defense research project Used deep learning to generate initial trajectories for optimization, 100x faster than prior approach	Agency for Defense Development (ADD) Mar 2021 – Jan 2023

ACTIVITIES

Auturbo Quadruped Robot Team Leader (Mar 2024 – Dec 2024), Regular Member (Mar 2023 – Dec 2023) Building StrideSim, a quadruped robot simulation platform on IsaacSim	Mar 2023 – Dec 2024
BARAM, Kwangwoon University Vice President (2019), Regular Member (2018, 2020-2021) Elected vice president over 60+ members; built and exhibited 6 robotic projects	Seoul, South Korea Mar 2018 – Feb 2021

HONORS

- Military Robotics Society Excellence Poster Award (Nov 2024)
- Ministry of National Defense Drone-Bot Challenge: Grand Prize, 2 consecutive years (2023, 2024)
- Open SW Mini Hackathon 3rd Prize (Nov 2020)
- Dean's List (Dec 2020, Jun 2019, Dec 2018)

SKILLS

Programming Languages: C++, Python

Robotics & Navigation: ROS 2, Nav2, SLAM (GLIM, DLIO), Behavior Trees, MPPI

Simulation & RL: MuJoCo, IsaacLab, PyTorch, mujlab

Tools & Infrastructure: Git, Docker, Jetson Orin, Ansible, Jenkins

Language Fluency: Intermediate-high English (TOEIC 810, OPIc IH), Native Korean